Back

Knowledge Check: The Stochastic Block Model and The Configuration Model

Practice Assignment • 30 min

Due Oct 21, 11:59 PM IST

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item

1. Why is the stochastic block model considered “stochastic”? 1 / 1 point

- Because it creates networks with random node labels
- Because it always creates random matrices P
- Because the edge creation is based on probabilistic rules
- Because the same model parameters always produce the same network

Nice work

Feedback: The model uses probabilistic edge creation, so networks can vary across runs.

2. What is the main advantage of the stochastic block model? 1 / 1 point

- It helps us simulate the dynamics of information diffusion
- It generates networks with community structure
- It produces networks with a power-law degree distribution
- It ensures each node has the same degree

Nice work

Feedback: The stochastic block model is designed to simulate networks that exhibit community structure—groups of nodes with denser internal connections.

3. What would most likely lead to a disassortative network in the stochastic block model? 1 / 1 point

- Equal probabilities for all entries in P
- High diagonal values in P
- A fully symmetric P matrix
- High off-diagonal values and low diagonal values in the matrix P

Nice work

Feedback: Disassortativity occurs when connections are more likely between different groups.

4. Which of the following statements about matrix P in the stochastic block model for undirected networks is true? 1 / 1 point

- It must be symmetric
- It can contain values greater than 1
- It determines the exact number of edges in the network
- The matrix includes weights for each edge

Nice work

Feedback: For undirected networks, the matrix P must be symmetric since P_ij and P_ji represent the same undirected edge probability.

5. Why does the friendship paradox occur in networks with positive degree variance? 1 / 1 point

- Because nodes are uniformly connected
- Because high-degree nodes are connected to low-degree nodes
- Because high-degree nodes appear more often as neighbors
- Because all nodes have equal access to information

Nice work

Feedback: Nodes with many connections are more likely to be encountered by traversing edges, making them overrepresented in neighbor samples.

6. Which of the following inputs are required to generate a network using the stochastic block model? 1 / 1 point

- [ ] A list of edge weights
- [ ] List of node degrees
- [x] Number of nodes

Nice work

Feedback: This is a fundamental input to define the size of the network.

- [x] A k×k symmetric matrix of probabilities

Nice work

Feedback: The kxk matrix defines both the within- and between-group connection probabilities.

7. In the stochastic block model, what does the diagonal of the matrix P represent? 1 / 1 point

- Probability of connecting nodes from different groups
- Probability of connecting nodes within the same group
- The strength of assortativity in the network
- Total number of nodes in the group

Nice work

Feedback: Diagonal values (e.g., P_11, P_22) represent the likelihood of internal group connections.

8. What must be true about the degree sequence used in the configuration model? 1 / 1 point

- All nodes must have the same degree
- Each degree must be a prime number
- The degree sequence must follow a Poisson distribution
- The sum of degrees must be an even number

Nice work

Feedback: Since each edge connects two nodes, the total number of stubs must be even.

9. What does the term “stub” refer to in the configuration model? 1 / 1 point

- A node with only one connection
- A network with no connections
- A placeholder for the adjacency matrix
- A half-edge waiting to be connected

Nice work

Feedback: A stub is an endpoint of an edge assigned to a node before connections are made.

10. Which of the following are characteristics of the stochastic block model? 1 / 1 point

- [ ] It uses directed edges by default
- [x] It can produce assortative and disassortative networks

Nice work

Feedback: By adjusting within- and between-group probabilities, different structures can be created.

- [x] It allows for modeling community structure

Nice work

Feedback: This is a key strength of the model.

- [ ] It guarantees equal degree across nodes

11. What unusual feature can occur in networks generated by the configuration model? 1 / 1 point

- Disconnected components of equal size
- Self-loops and parallel edges
- Cycles of exactly three nodes
- Directional edges between each node pair

Nice work

Feedback: These features can arise due to random stub pairings in the configuration model.

12. Why is the configuration model useful in studying real-world networks? 1 / 1 point

- [ ] It models networks with hierarchical edge weights
- [x] It helps analyze how degree distribution affects network processes

Nice work

Feedback: Researchers can simulate diffusion or dynamics on networks with controlled structures.

- [x] It allows control over node degrees for network generation

Nice work

Feedback: This makes the model valuable for testing hypotheses about degree effects.

- [ ] It creates perfect simulations of real-world networks